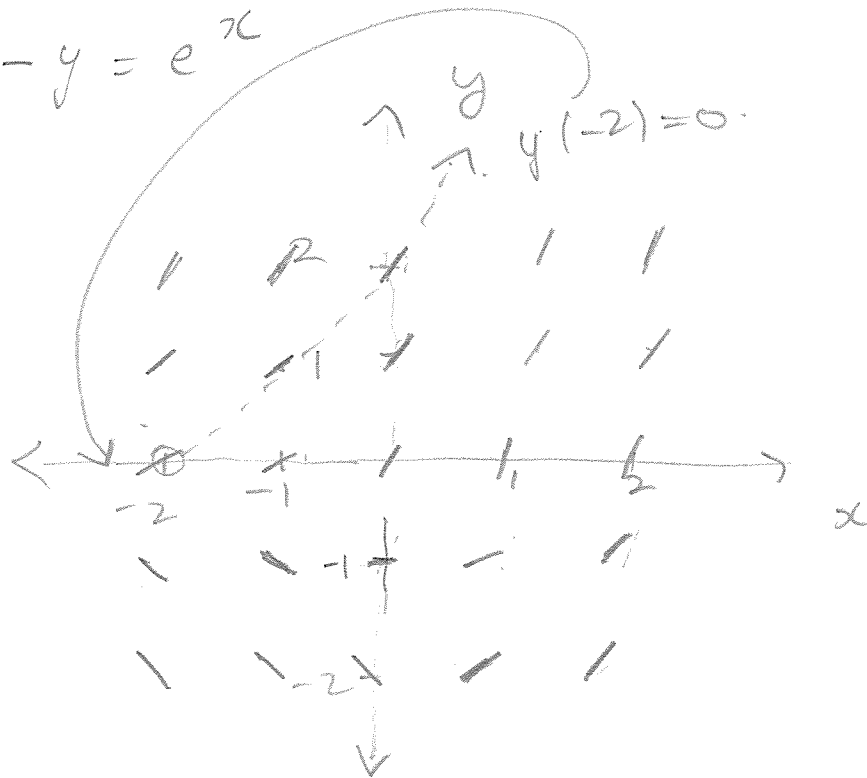


Tutorial 2.

1. $y' - y = e^x$



	-2	-1	0	1	2
+2	$e^{-2} + 2$	$e^{-1} + 2$	$1 + 2 = 3$	$e + 2$	$e^2 + 2$
+1	$e^{-2} + 1$	$e^{-1} + 1$	$1 + 1 = 2$	$e + 1$	$e^2 + 1$
0	e^{-2}	e^{-1}	$e^0 = 1$	e^1	e^2
-1	$e^{-2} - 1$	$e^{-1} - 1$	$1 - 1 = 0$	$e^1 - 1$	$e^2 - 1$
-2	$e^{-2} - 2$	$e^{-1} - 2$	$1 - 2 = -1$	$e^1 - 2$	$e^2 - 2$

Full in y'
 $y' = e^x + y$

$$a. a) y' = 2(y-1)$$

$$\frac{y'}{y-1} = 2.$$

$$\int \frac{dy}{y-1} = \int 2 dx$$

$$\ln(y-1) = 2x + C$$

$$e^{2x+C} = y-1$$

$$y = Ce^{2x} + 1$$

$$b) y' = 4y$$

$$\int \frac{dy}{y} = \int 4 dx$$

$$\ln y = 4x + C$$

$$e^{4x+C} = y$$

$$y = Ce^{4x}$$

$$c) y' + xy = x$$

$$y' = x - xy = x(1-y)$$

$$\frac{y'}{1-y} = x$$

$$\int \frac{dy}{1-y} = \int x dx$$

$$-\ln(1-y) = \frac{1}{2}x^2 + C$$

$$\ln(1-y) = -\frac{1}{2}x^2 - C$$

$$e^{-\frac{1}{2}x^2 - C} = 1-y$$

$$y = Ce^{-\frac{1}{2}x^2} + 1$$

$$d) xy' = x^3 y \quad (x \neq 0)$$

$$\frac{y'}{y} = x^2$$

$$\int \frac{dy}{y} = \int x^2 dx$$

$$\ln y = \frac{1}{3}x^3 + C$$

$$e^{\frac{1}{3}x^3 + C} = y$$

$$y = Ce^{\frac{1}{3}x^3}$$

Tutorial 2 dtd.

3. a) $y' = x + y$

First order

Linear.

$$y' - y = x$$

Nonhomogeneous

use integrating factor $\mu(x)$

$$y' + p(x)y = r(x)$$

Here $p(x) = -1$ and $r(x) = x$

$$\mu(x) = e^{\int p(x) dx} = e^{-x}$$

$$\text{So } \mu(x)y(x) = \int \mu(x)r(x)dx + C$$

$$e^{-x}y(x) = \int e^{-x}x dx + C$$

$\int e^{-x} dx$, integrate by parts

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

with $u(x) = x$ and $\frac{dv}{dx} = e^{-x} dx$

then $\frac{du}{dx} = 1$ and $v = -e^{-x}$ (up to constant)

$$\begin{aligned} \text{so } \int e^{-x} dx &= -xe^{-x} - \int -e^{-x} dx \\ &= -xe^{-x} - e^{-x} + C \end{aligned}$$

$$\begin{aligned} \text{Hence } y(x) &= e^x [-xe^{-x} - e^{-x} + C] \\ &= -x - 1 + ce^x \end{aligned}$$

$$3b) y' + 2y = x^2 + 4x + 7$$

use integrating factor

$$p(x) = 2, \quad r(x) = x^2 + 4x + 7$$

$$\mu(x) = e^{\int p(x) dx} = e^{2x}$$

(replaced.)

$$\text{so } e^{2x} y(x) = \int e^{2x} (x^2 + 4x + 7) dx + C$$

(three integrals here.)
 I_2, I_1, I_0

Integrating by parts

$$\int e^{2x} x^2 dx = \frac{1}{2} e^{2x} x^2 - \int 2x \left(\frac{1}{2} e^{2x} \right) dx$$

$$u(x) = x^2 \quad \frac{du}{dx} = 2x$$

$$\frac{dv}{dx} = e^{2x}$$

$$v = \frac{1}{2} e^{2x}$$

(This is I_2)

$$i) \text{ if } z = 2x \text{ then } \int \frac{z}{2} e^z \frac{dz}{2} = \int \frac{z}{2} e^z dx$$

$$\int z e^z dz = z e^z - \int e^z dz = z e^z - e^z + C_1$$

$$v(z) = e^z \quad u(z) = z$$

$$\frac{dv}{dz} = e^z$$

part of I_2

$$\text{so } \int e^{2x} x^2 dx = \frac{1}{2} e^{2x} x^2 - \frac{1}{4} \left[2x e^{2x} - e^{2x} + C_1 \right]$$

$$ii) \int x e^{2x} dx = \int \frac{w}{2} e^w \frac{dw}{2} \quad w = 2x$$

$$= \frac{1}{4} \int w e^w dw = \frac{1}{4} (w e^w - e^w + C_2)$$

$$= \frac{1}{4} 2x e^{2x} - \frac{e^{2x}}{4} + C_2$$

Tutorial 2

3b) continued.

(replaced)

$$\text{iii). } \int e^{2x} \cdot 7 dx = \frac{7}{2} e^{2x} + c_3$$

$$\text{so } e^{2x} y(x) = \frac{1}{2} e^{2x} x^2 - \frac{1}{4} \left[2x e^{2x} - e^{2x} + c_1 \right]$$

$$+ 4 \left[\frac{1}{4} 2x e^{2x} - \frac{e^{2x}}{4} + \frac{c_2}{4} \right]$$

$$+ \frac{7}{2} e^{2x} + c_3$$

$$\text{so } y(x) = \frac{x^2}{2} - \frac{x}{2} + \frac{1}{4} + (c_1 e^{-2x}) \left(-\frac{1}{4}\right)$$

$$+ 2x - 1 + c_2 e^{-2x}$$

$$+ \frac{7}{2} + c_3 e^{-2x}$$

$$= \frac{x^2}{2} + \frac{3x}{2} + \frac{11}{4} + C e^{-2x}$$

This is messy

→ simpler order of calculation follows.

3 b) (simpler method).

$$y' + 2y = x^2 + 4x + 7$$

use integrating factor

$$p(x) = 2, \quad r(x) = x^2 + 4x + 7$$

$$\mu(x) = e^{\int 2 dx} = e^{2x}$$

$$\text{so } e^{2x} y(x) = \int e^{2x} (x^2 + 4x + 7) dx + C$$

Work out $\int e^{2x} x dx$ first by parts.

$$\frac{dV}{dx} = e^{2x}$$

$$\text{so } \int e^{2x} x dx$$

$$V = \frac{e^{2x}}{2}, \quad u = x$$

$$= \frac{x}{2} e^{2x} - \int \frac{e^{2x}}{2} dx$$

$$= \frac{x}{2} e^{2x} - \frac{e^{2x}}{4} + C_1$$

$$\text{Next } \int x^2 e^{2x} dx = \frac{x^2}{2} e^{2x} - \int 2x \left(\frac{1}{2} e^{2x} \right) dx$$

$$\frac{dV}{dx} = e^{2x}$$

$$= \frac{x^2}{2} e^{2x} - \int x e^{2x} dx$$

$$V = \frac{1}{2} e^{2x}$$

$$u = x^2$$

$$= \frac{x^2}{2} e^{2x} - \left[\frac{x}{2} e^{2x} - \frac{e^{2x}}{4} + C_1 \right]$$

$$\int e^{2x} dx = \frac{1}{2} e^{2x} + C_2$$

$$e^{2x} y(x) = \frac{x^2}{2} e^{2x} - \int x e^{2x} dx + 4 \int x e^{2x} dx + 7 \int e^{2x} dx$$

$$= \frac{x^2}{2} e^{2x} + 3 \int x e^{2x} dx + \frac{7}{2} e^{2x} + C_1$$

$$= \frac{x^2}{2} e^{2x} + 3 \left[\frac{x}{2} e^{2x} - \frac{e^{2x}}{4} + C_1 \right] + \frac{7}{2} e^{2x} + C_2$$

$$y(x) = \frac{x^2}{2} + \frac{3x}{2} - \frac{3}{4} + \frac{14}{4} + C e^{-2x}$$

$$= \frac{x^2}{2} + \frac{3x}{2} + \frac{11}{4} + C e^{-2x}$$

Tutorial 2. ctd.

$$3 \text{ c). } xy' + 4y = 2x$$

$$y' + \frac{4}{x}y = 2 \quad x \neq 0.$$

first order, linear, nonhomogeneous
use integrating factor.

$$p(x) = \frac{4}{x}, \quad r(x) = 2.$$

$$\mu(x) = e^{\int p(x) dx} = e^{\int \frac{4}{x} dx} = e^{4 \ln x} = x^4$$

$$\text{so } x^4 y(x) = \int x^4 \cdot 2 dx + C$$

$$x^4 y(x) = \frac{2}{5} x^5 + C$$

$$y(x) = \frac{2x}{5} + Cx^{-4}$$

$$3d) \quad xy' - 4y = x^5 e^x$$

$$y' - \frac{4}{x} y = x^4 e^x \quad x \neq 0$$

$$p(x) = -\frac{4}{x}, \quad r(x) = x^4 e^x$$

$$\int -\frac{4dx}{x}$$

$$\text{integrating factor } \mu(x) = e$$

$$= e^{-4 \ln x} = x^{-4}$$

$$\text{so } x^{-4} y(x) = \int x^{-4} (x^4 e^x) dx + C$$

$$x^{-4} y(x) = e^x + C$$

$$y(x) = x^4 e^x + C x^4$$

Tutorial 2 ctd. Initial Value Problems

4 a) $y' = yx^2$ $y(0) = 1$

$$\frac{y'}{y} = x^2 \quad \text{Separable first order ODE}$$

$$\int \frac{dy}{y} = \int x^2 dx$$

$$\ln y = \frac{1}{3} x^3 + C$$

$$e^{\frac{1}{3}x^3 + C} = y = C e^{\frac{x^3}{3}}$$

Sub $x=0, y=1$; $C=1$; so $y = e^{\frac{x^3}{3}}$

b) $xy' - y^2 = 0$ $y(0) = 2$

if $x=0$, this is satisfied, also

$$y' = y^2$$

$$\int \frac{dy}{y^2} = \int dx$$

$$-y^{-1} = x + C$$

$$\frac{1}{y} = -x - C$$

$$y = \frac{1}{-x + C}$$

if $x=0$

$$y = \frac{1}{C} = 2$$

so $C = \frac{1}{2}$

$$y = \frac{1}{-x + \frac{1}{2}}$$

$$= \frac{1}{\frac{1}{2} - x}$$

$$4c). y' - 2y = e^x \quad y(0) = 0$$

$$\text{use } p(x) = -2, \quad r(x) = e^x$$

$$\text{integrating factor } \mu(x) = e^{\int -2 dx} = e^{-2x}$$

$$\text{Then } e^{-2x} y(x) = \int e^{-2x} e^x dx + C = -e^{-x} + C$$

$$\text{so } y(x) = -e^x + C e^{2x}$$

$$\text{if } x=0 \text{ then } y=0 \text{ so } 0 = -1 + C$$

$$\text{so } C = 1 \text{ and } y(x) = -e^x + e^{2x}$$

$$d). xy' + y = x^2 \quad y(1) = 0 \quad \left(\text{if } x=0 \text{ then } y=0 \right)$$

$$y' + \frac{y}{x} = x \quad \text{otherwise}$$

$$p(x) = \frac{1}{x}, \quad r(x) = x, \quad \mu(x) = e^{\int \frac{dx}{x}} = e^{\ln x} = x$$

$$\text{so } xy(x) = \int x^2 dx + C = \frac{1}{3} x^3 + C$$

$$y(x) = \frac{x^2}{3} + \frac{C}{x}$$

$$\text{if } x=1$$

$$y(1) = \frac{1}{3} + C = 0$$

$$\text{so } C = -\frac{1}{3}$$

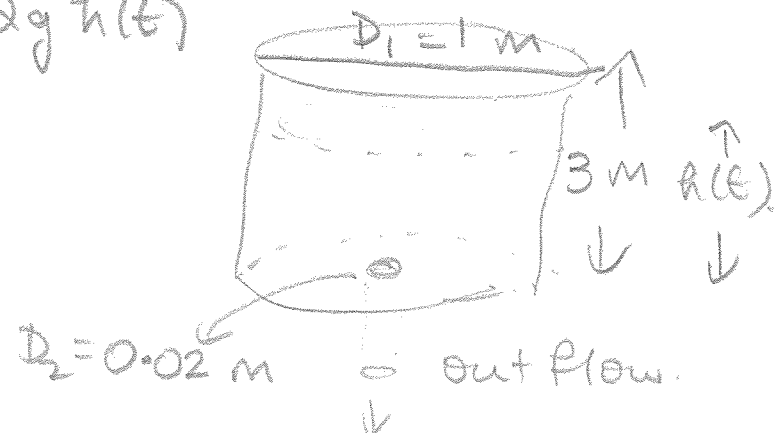
$$y(x) = \frac{x^2}{3} - \frac{1}{3x} = \frac{1}{3} \left(x^2 - \frac{1}{x} \right)$$

Tutorial 2. dtd.

5. Water flowing through a tank: Torricelli's law

$$v(t) = 0.6 \sqrt{2gh(t)}$$

$$g = 9.8 \text{ m/s}^2$$



Rate of change of volume Vol.

$$\frac{d\text{Vol}}{dt} = (\text{large surface area}) \frac{dh}{dt} (-1)$$

$$= (\text{Small surface area}) v(t)$$

(Conservation of mass or volume)
as density is constant

$$\text{so } \pi D_1^2 \left(-\frac{dh}{dt} \right) = \pi D_2^2 0.6 \sqrt{2gh(t)}$$

$$\frac{dh}{dt} = -\frac{D_2^2}{D_1^2} 0.6 \sqrt{2g} (h(t))^{\frac{1}{2}}$$

$$= -(0.02)^2 \times 0.6 (\sqrt{2g}) h(t)^{\frac{1}{2}}$$

$$= -1.06253 \times 10^{-3} (h(t))^{\frac{1}{2}}$$

$$\int \frac{dh}{h^{\frac{1}{2}}} = -1.06253 \times 10^{-3} \int dt$$

$$= -1.06253 \times 10^{-3} t + C$$

$$2h^{\frac{1}{2}} = -1.06253 \times 10^{-3} t + c$$

$$h^{\frac{1}{2}} = \frac{-1.06253 \times 10^{-3} (t)}{2} + \frac{c}{2}$$

when $t=0$, $h=3$

$$\sqrt{3} = \frac{c}{2}$$

$$h^{\frac{1}{2}} = -\frac{1.06253 \times 10^{-3} t}{2} + \sqrt{3}$$

$$y_h^{\frac{1}{2}} = -5.31 \times 10^{-4} t + \sqrt{3}$$

$$y_h = \left(-5.31 \times 10^{-4} t + \sqrt{3} \right)^2$$

Tutorial 2 ctd.

6. Electromotive force (EMF) Volts.

$$R = 1 \Omega, \quad L = 1 \text{ H. (Henry).}$$

$$E(t) = \sin(t)$$

$$LI' + RI = E$$

$$I' + \frac{R}{L} I = \frac{E}{L} = \frac{\sin(t)}{L} \text{ simplifies to}$$

$$I' + I = \sin(t) \quad \text{solve for } I(t)$$

$$I' + p(t)I = r(t)$$

$$p(t) = 1, \quad r(t) = \sin(t).$$

$$\mu(t) = e^{\int p(t) dt} = e^t$$

$$e^t I(t) = \int e^t \sin t \, dt + C$$

$$\text{let } u = e^t, \quad \text{let } \frac{dv}{dt} = \sin t$$
$$\frac{du}{dt} = e^t \quad v = -\cos t$$

$$\int e^t \sin t \, dt = -e^t \cos t - \int -\cos t e^t \, dt$$
$$= -e^t \cos t + \int \cos t e^t \, dt.$$

now choose $\frac{dv}{dt} = \cos t$ $v = \sin t$ $u(t) = e^t$

$$\rightarrow -e^t \cos t + \left[(\sin t) e^t - \int \sin t e^t \, dt \right]$$

$$\text{so } 2 \int e^t \sin t \, dt = -e^t \cos t + (\sin t) e^t + 2C$$

$$\text{so } \int e^t \sin t \, dt = \frac{1}{2} \left[\sin t e^t - \cos t e^t \right] + C_1$$

$$\text{so } e^t I(t) = \frac{1}{2} [\sin t e^t - \cos t e^t] + C$$

$$I(t) = \frac{\sin t}{2} - \frac{\cos t}{2} + C e^{-t}$$

$$\text{if } t=0, I(0)=0$$

$$\text{so } 0 = 0 - \frac{1}{2} + C, \text{ so } C = \frac{1}{2}$$

$$I(t) = \frac{\sin t}{2} + \frac{\cos t}{2} + \frac{e^{-t}}{2}$$



oscillating.



decaying.

7. Value $\frac{dV(t)}{dt} = -r(10-t)$ $r > 0$.

$$V(t) = -\frac{r}{2}(10-t)^2 + C$$

$$V(t) = \frac{r}{2}(10-t)^2 + C \quad 0 \leq t \leq 10$$

when $t = 0$, $V(0) = 5000$

$$\textcircled{1} \quad 5000 = \frac{r}{2} \times 100 + C = 50r + C$$

when $t = 1$, $V(1) = 3800$.

$$3800 = \frac{r}{2}(9)^2 + C$$

$$\textcircled{2} \quad 3800 = \frac{81r}{2} + C$$

$$\text{so } 1200 = \left(\frac{100-81}{2} \right) r = \frac{19r}{2}$$

$$r = \frac{2400}{19} = 126.316$$

and $C = 5000 - 50r = 1315.789$

$$V(t) = 63.16(10-t)^2 - 1316$$

P.T.O.

$$\text{or } V(t) = \frac{\Gamma}{2} (10-t)^2 + C$$

$$= \frac{\Gamma}{2} (100 - 20t + t^2) + C$$

$$= 50\Gamma + C + \Gamma \left(-10t + \frac{t^2}{2} \right)$$

$$= 5000 - \Gamma \left(10t - \frac{t^2}{2} \right)$$

$$= 5000 - 126 \left(10t - \frac{t^2}{2} \right)$$